Numerical Problems on Binomial, Poisson and Normal Probability Distributions.

1. Define a binomial variate and derive the expressions for its mean and variance.

A fair die is tossed thrice. A success is getting 1 or 6 on a toss. Find the mean and variance of the number of successes.

2. An underground mine has 5 pumps installed for pumping out storm water. The probability of any one of the pumps failing during the storm is 1/8. What is the probability that (i) at least 4 pumps (ii) all the pumps will be working during a particular storm?

3. If the probability is 0.15 that any person will dislike the taste of a new toothpaste, what is the probability that at least three persons of 7 randomly selected persons will dislike it? Calculate the mean and variance of the concerned probability distribution.

4. Show Poisson distribution is the limiting form of a binomial distribution.

An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year?

5. If the probability that an individual suffers a bad reaction from injection of a given serum is 0.001, determine the probability that out of 2000 individuals (a) exactly three (b) at least three and (c) not more than two individuals will suffer a bad reaction.

6. Under what conditions Poisson probabilities are good approximation of binomial probabilities?

The probability of fatal accident in an explosive industry employing 300 workers during the year is 0.005. If 1000 such industries are in the existence, find the number of industries in which there will be (i) no fatal accident (ii) at least two fatal accidents and (iii) not more than two fatal accidents in the year.

7. The probability of fatal accident in an explosive industry employing 250 workers during the year is 0.01. If 100 such industries are in the existence, find the number of industries in which there will be (i) at least two fatal accidents and (ii) not more than two fatal accidents in the year.

8. Assuming that the diameters of 1000 brass plugs taken consecutively from a machine form a normal distribution with mean 0.7515 cm and standard deviation 0.0020 cm, how many of the plugs are likely to be rejected if the approved diameter is 0.752 ± 0.004 cm?

9. A manufacturer knows from experience that the resistance of resistors he produces is normal with mean 100 ohms and standard deviation 2 ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms?
Given $P(Z > 1) = 0.1587$, where Z is standard normal variate.

10. Define standard normal variate and write down its probability density function.
The life of army shoes is normally distributed with mean 8 months and standard deviation 2 months. If 5000 pairs are issued how many pairs would be expected to need replacement after 12 months? Given $P(Z \geq 2) = 0.0228$, Z is standard normal variate.

11. The weekly wages of 200 workers in a factory are normally distributed with a mean of Rs. 200.00 and a variance of Rs. 400.00. Estimate the lowest weekly wages of the 197 highest paid workers and the highest weekly wages of the 197 lowest paid workers.

12. Incandescent electric bulbs of a certain make have a mean life of 800 hours with standard deviation of 150 hours. Out of fifty thousand bulbs of this make used for street lighting, how many of them would fuse in first 650 hours, if the distribution of the life on bulbs is assumed to be normal? On the same assumption, how many bulbs would still be burning after 950 hours?

13. Define the normal and standard normal variates. Write down their respective pdfs.
The inside diameter of a piston ring is normally distributed with mean of 12 cm and a standard deviation of 0.02 cm. (i) What fraction of the piston rings will have diameters exceeding 12.05cm? (ii) What inside diameter value k has a probability of 0.90 that being exceeded? What is the probability that the inside diameter will fall between 11.95 cm and 12.05 cm?

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16. Define the normal and standard normal variates. Mention three important characteristics of a normal distribution.

A random variable has a normal distribution with variance 100. If the probability is 0.8212 that it will take on a value less than 82.5, what is the probability that it will take on a value greater than 58.3?